Activity 2 — Employee Payroll and Deduction Analysis

1. Problem Statement

Develop a Python program to calculate an employee's gross salary using basic pay, allowances, overtime and incentives. Calculate deductions and display the final net salary.

2. Objectives

- To calculate gross salary.
- To calculate salary deductions.
- To determine net salary.
- To understand variables, arithmetic operators and conditional statements.

3. Assumptions

For this example, assume:

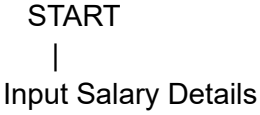
- Basic salary = ₹20,000
- Allowances = ₹3,000
- Overtime pay = ₹1,000
- Incentives = ₹500
- Deduction rate = 5% of gross salary

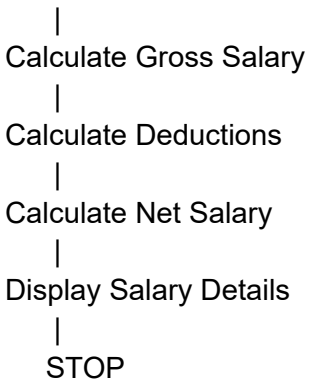
These are sample values for educational purposes.

4. Algorithm

1. Start.
2. Read the basic salary, allowances, overtime pay and incentives.
3. Calculate gross salary by adding all earnings.
4. Calculate deductions as 5% of gross salary.
5. Calculate net salary by subtracting deductions from gross salary.
6. Display gross salary, deductions and net salary.
7. Stop.

5. Flowchart





6. Pseudocode

```
START
INPUT basic_salary
INPUT allowances
INPUT overtime
INPUT incentives

gross_salary = basic_salary + allowances
             + overtime + incentives

deductions = gross_salary * 0.05
net_salary = gross_salary - deductions

PRINT gross_salary
PRINT deductions
PRINT net_salary
STOP
```

7. Python Source Code

File name: `activity4.py`

```
basic_salary = float(input("Enter basic salary: "))
allowances = float(input("Enter allowances: "))
overtime = float(input("Enter overtime pay: "))
incentives = float(input("Enter incentives: "))

if basic_salary < 0 or allowances < 0 or overtime < 0 or incentives < 0:
    print("Invalid salary details")
else:
    gross_salary = basic_salary + allowances + overtime + incentives

    deductions = gross_salary * 0.05
    net_salary = gross_salary - deductions
```

```
print("Gross Salary: Rs.", gross_salary)
print("Deductions: Rs.", deductions)
print("Net Salary: Rs.", net_salary)
```

8. Test Cases

Basic Salary	Allowances	Overtime	Incentives	Expected Net Salary
20000	3000	1000	500	₹23,275
15000	2000	500	500	₹17,100
-1000	2000	500	500	Invalid salary details

9. Output Screenshot and Conclusion

Sample Output:

Enter basic salary: 20000
Enter allowances: 3000
Enter overtime pay: 1000
Enter incentives: 500
Gross Salary: Rs. 24500.0
Deductions: Rs. 1225.0
Net Salary: Rs. 23275.0

Output Screenshot: Run `activity4.py` in Python IDLE or VS Code and capture a screenshot showing the inputs and output. Save it as `activity4_output.png`.

Conclusion: The program calculates gross salary, deductions and net salary using Python arithmetic operations and conditional statements.

Input code:

```
basic_salary = float(input("Enter basic salary: "))
allowances = float(input("Enter allowances: "))
overtime = float(input("Enter overtime pay: "))
incentives = float(input("Enter incentives: "))

if basic_salary < 0 or allowances < 0 or overtime < 0 or incentives < 0:
    print("Invalid salary details")
else:
    gross_salary = basic_salary + allowances + overtime + incentives

    deductions = gross_salary * 0.05
    net_salary = gross_salary - deductions

    print("Gross Salary: Rs.", gross_salary)
    print("Deductions: Rs.", deductions)
    print("Net Salary: Rs.", net_salary)
```

Output:

```
Enter basic salary: 2000
Enter allowances: 3000
Enter overtime pay: 1000
Enter incentives: 500
Gross Salary: Rs. 6500.0
Deductions: Rs. 325.0
Net Salary: Rs. 6175.0

========== RESTART:
Enter basic salary: 15000
Enter allowances: 2000
Enter overtime pay: 500
Enter incentives: 500
Gross Salary: Rs. 18000.0
Deductions: Rs. 900.0
Net Salary: Rs. 17100.0

========== RESTART:
Enter basic salary: -1000
Enter allowances: 2000
Enter overtime pay: 500
Enter incentives: 500
Invalid salary details
```